Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/20/2005
PI Name:	Weichman, Peter B Ph.D.		
Project Title:	Theoretical investigations of equilibrium and nonequilibrium superfluid dynamics with application to future ground-based and microgravity experiments		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Burlington	**State:**	MA
Zip Code:	01803-5024	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	GSN 4497		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity		

Task Description:	Various aspects of the theory of equilibrium and nonequilibrium superfluidity will be considered with special emphasis on properties relevant to proposed and planned future microgravity experiments, and their ground-based precursors. Of special interest are steady equilibrium and nonequilibrium interfacial states, in which gravity (equilibrium) and heat current (nonequilibrium), or a combination of the two, supports coexistence of superfluid and normal phases with a well defined interface between them. Approximate mean field and (where feasible) more accurate renormalization group calculations will be performed (a) to extend present understanding of the steady state structure of these interfaces, and (b) to understand their near-steady state dynamics under the influence of various experimental probes, such as second sound. Also of interest are further investigations into the nature of the so-called self-organized critical (SOC) state, which supports a finite temperature gradient in the superfluid phase by generating a stream of vortex lines propagating across the direction of heat flow. I will investigate the fluctuations and spatial correlations in this state, its associated nonequilibrium specific heat, and the effects of a superfluid-SOC interface on measurements of the heat capacity at constant heat current.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This research grant was initiated only in August, 2004, and the initial funding was only for about 2.5 months of PI labor, of which approximately half has been spent to date. I have begun development and analysis of the renormalization group (RG) theory of the gravity-induced equilibrium superfluid-normal fluid interface. The work is still in progress, so a formal write-up has not yet been prepared. The essential results to date are summarized in the Annual Report.
Bibliography Type:	Description: (Last Updated: 01/12/2021)
Articles in Peer-reviewed Journals	Radzihovsky L, Park J, Weichman PB. "Superfluid transitions in bosonic atom-molecule mixtures near a Feschbach resonance." Phys Rev Lett. 2004 Apr 23;92(16):160402. https://doi.org/10.1103/PhysRevLett.92.160402 ; PMID: 15169205 , Apr-2004